EIC and ePIC: the machine and the detector this programme is projected on

2026-08-27 · the specification behind reports 1, 2 and 4 · every row carries its source and its **status** — design value, requirement, measured, derived here, or unknown · figures from `evgen/scripts/eic_beam_figures.py`

Every projection in this programme rests on numbers that belong to the accelerator or to the detector, not to us. This note gathers them in one place, says where each comes from, and — more importantly — says what each one *is*: a design value in a conceptual design report, a requirement the detector must meet, something actually measured, or a gap we have filled with a stand-in. The lithium case is called out separately, because almost nothing about it is specified and the little that is has to be derived.

> **Three things a reader should take away.** (1) **Ion energies are set by speed, not rigidity.** The two rings must share a revolution period, so the hadron γ is fixed by the ring circumference and the magnets supply whatever rigidity that demands. The ^{6}Li menu is **41, and 99–138 GeV/u**, with nothing between (§1). (2) **Almost nothing about a lithium beam is published.** Not the emittance, not the intensity, not β*, not a luminosity. What can be derived is derived in §4; what cannot is listed in §6 with an owner. (3) **This programme's detector model is partly stand-in.** The tracking resolution and the electron-ID curve in `polligen.reco` carry no published ePIC source, and §6 says so explicitly rather than letting them pass as specification.

1 · The collider, and why the ion energies are what they are

The EIC is two rings in the RHIC tunnel: the **Electron Storage Ring** at 5, 10 and 18 GeV, and the **Hadron Storage Ring** carrying protons to 275 GeV and ions to the same magnetic rigidity. ePIC sits at IP6; a second interaction region at IP8 is designed but not funded. The crossing angle is 25 mrad at IP6 and 35 mrad at IP8, recovered by crab cavities.

> **The constraint that sets the ion menu.** Colliding bunches requires the two rings to have the *same revolution period*. The electrons are ultrarelativistic at every energy, so their period is fixed; the hadrons must match it, and since the circumference is adjustable only within a narrow range, **the hadron β — hence γ — is quantised by the ring geometry**. The magnets then have to supply whatever rigidity that γ implies for the species in the ring, up to the 275 GeV proton rigidity of 917.3 T·m.
>
> So two different limits bind at two different ends. At the *top* the magnets bind and the ion is **rigidity-capped** at p/nucleon = 275 GeV × (Z/A). At the *bottom* the circumference binds and the ion is **γ-matched** — the same *speed* as the proton configuration, and therefore a different energy per nucleon than rigidity scaling would give.

The check that settles it. Yellow Report Table 10.2 lists gold at **41 GeV/u** (γ = 44.02) against the 41 GeV proton's γ = 43.70 — equal to 0.7%, which is exactly the u versus m_p mass difference. Rigidity scaling would have put gold at 41 × 79/197 = 16.4 GeV/u. It does not. The same rule reproduces the published ^{3}He menu — *"41, and 100–183 GeV/nucleon"* — exactly.

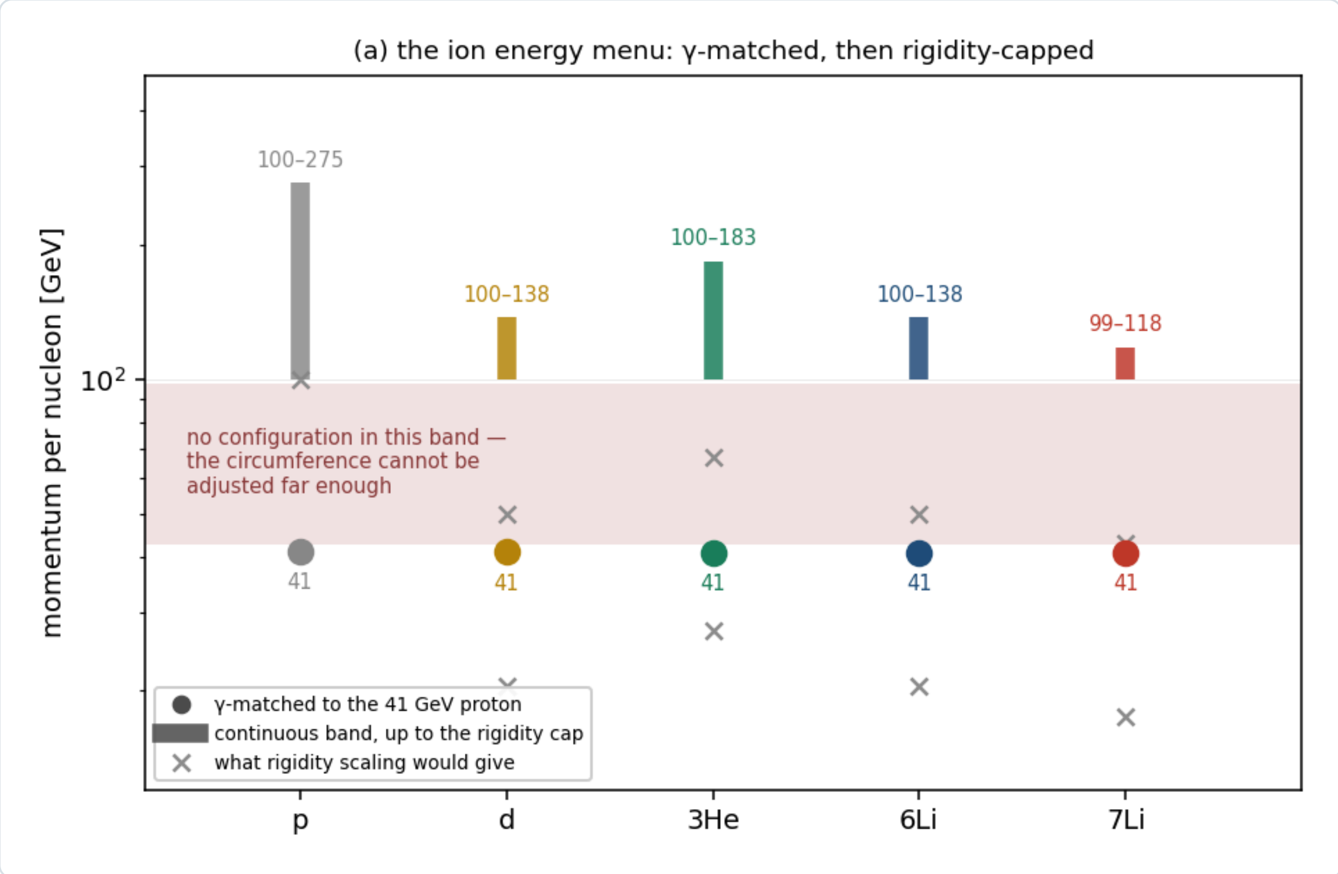

Figure 1 · The ion energy menu. Each species has an isolated γ-matched point at ≈41 GeV/u and a continuous band from ≈99 GeV/u up to its own rigidity cap — and nothing in between, because the circumference cannot be adjusted far enough. Crosses mark what rigidity scaling would have given, which is what this repository used until 2026-08-27. `eic_beam_figures.py`.

SPECIES	A/Z	Γ-MATCHED POINT	CONTINUOUS BAND	RIGIDITY CAP	STATUS
p	1	41 GeV	100–275 GeV	275 GeV	design value
d	2	41.0 GeV/u	100–137.5	137.5	design value
^{3}He	3/2	40.9	99.8–183.3	183.3	design value (menu published)
^{6}Li	2	**40.8**	**99.5–137.5**	137.5	**derived**; top ≈138 in EPIOS
^{7}Li	7/3	**40.8**	**99.5–117.9**	117.9	**derived**; top ≈117 in EPIOS
Au	2.49	41	– (110 published)	110.3	design value

Table 1 — `beams.default_configs` computes these; running the same machinery on a *proton* returns 41 / 100 / 275 exactly, which is the implementation's self-check. Only the lithium rows are derived rather than published, and only their top energies appear in EPIOS.

2 · The electron beam

QUANTITY	VALUE	SOURCE	STATUS
energies	5, 10, 18 GeV	YR §10.1	design value
β* (h/v)	45 / 5.6 cm	EPIOS Table I (EIC column)	design value
RMS emittance (h/v)	20 / 1.3 nm	EPIOS Table I	design value
bunch charge	28 nC	EPIOS Table I	design value
bunches	1160	EPIOS Table I	design value
divergence h/v, 18 GeV	202 / 187 (HD), 89 / 82 μrad (HA)	YR Table 10.1	design value
Δp/p	5.8–10.9 ×10^{-4}	YR Table 10.1	design value
polarization	70% assumed in the YR physics studies	YR §10.1	assumption for projections

3 · The hadron beam

Yellow Report **Table 10.1** (e+p) and **10.2** (e+Au) are the authoritative divergence and momentum-spread tables, and they carry both optics configurations. The **high-divergence** and **high-acceptance** settings differ by a β* choice: the YR states explicitly that $\sigma \propto 1/\sqrt{\beta^*}$ and $\beta^* \propto 1/L$, so acceptance is bought with luminosity.

PROTON CONFIG	HD H/V [MRAD]	HA H/V	ΔP/P [10^{-4}]	L [10^{33} CM^{-2}S^{-1}] HD / HA
275 GeV, 290 bunches	150 / 150	65 / 65	6.8	1.54 / 0.32
275 GeV, 1160 bunches	119 / 119	65 / 65	6.8	10.0 / 3.14
100 GeV	220 / 220	180 / 180	9.7	4.48 / 3.14
100 GeV (vs e 5)	206 / 206	180 / 180	9.7	3.68 / 2.92
41 GeV	220 / 380	**220 / 380**	10.3	0.44 / 0.44

Table 2 — YR Table 10.1, hadron rows, verbatim. **Status: design value.** Note the last row: at 41 GeV the two optics are *identical* — the only configuration with no separate high-acceptance option, and the one where this programme most needs one (§6, and report 4 §7).

GOLD CONFIG	STRONG HADRON COOLING H/V	ΔP/P	STOCHASTIC COOLING H/V	ΔP/P
110 GeV/u	218 / 379 µrad	6.2×10^{-4}	77 / 380	10×10^{-4}
41 GeV/u	275 / 377	10×10^{-4}	174 / 302	13×10^{-4}

Table 3 — YR Table 10.2, gold rows. The two cooling scenarios differ by a factor 2.8 horizontally at 110 GeV/u, which is a reminder that the cooling choice is worth as much as the optics choice.

Other hadron parameters (EPIOS Table I, EIC column, design values): β* = 80 / 7.2 cm, RMS emittance 11 / 1 nm, bunch charge 11.0 nC, 1160 bunches, bunch length 6 cm, $\sqrt{s}$ = 105 GeV, peak luminosity 10^{37} cm^{-2} s^{-1}. The beam is deliberately **flat**, $\varepsilon_x/\varepsilon_y \approx 11$, with β* matched to it so that the two *divergences* come out nearly equal — which is why Table 2's h and v agree at 275 and 100 GeV but not at 41.

4 · Lithium: published, derived, unknown

What is published is thin. EPIOS (arXiv:2510.10794, PRC 113:060501) gives the spin factors and the top energies and states the case; it gives *no* lithium luminosity, emittance or intensity, and describes the polarized-lithium source as early-stage. Hamwi & Hoffstaetter (arXiv:2509.18558, PRAB 29:073501) give the resonance spectrum.

QUANTITY	^{6}LI	^{7}LI	SOURCE / STATUS
spin	1	3/2	nuclear data
A/Z	2	7/3	—
G factor	−0.178 (EPIOS), −0.1818 (Hamwi Table I)	+1.532 / +1.5196	published; the two disagree in the last digits
max |Gγ|, resonances	27, 81	191, 573	Hamwi Table I, **design study**
snake scheme	deuteron-like: **not amenable to Siberian snakes** (small G needs impractically long snakes); partial solenoid + tune jumps	partial snakes	Hamwi; EPIOS
top energy	137.5 GeV/u	117.9 GeV/u	derived from the rigidity cap; ≈138 / ≈117 in EPIOS
energy menu	41, and 99–138	41, and 99–118	**derived** (§1)
σ_θ h/v, high acceptance	220/380, 180/180, 92/92 µrad	≈same at the two lower	**derived** (below)
Δp/p	7–10 ×10^{-4}	same	**derived**; species-insensitive

emittance, intensity, β*, luminosity	— nothing published —	**unknown** (§6)

4.1 · The divergence, derived

$\sigma_\theta = \sqrt{\varepsilon_N/(\beta\gamma\cdot\beta^*)}$, so at a given machine configuration — where β* is common to every species because the lattice is set by rigidity — the only species dependence is through βγ. And §1 says a **γ-matched** ion has the proton's βγ exactly. So:

$$\sigma_\theta(\text{ion}) = \sqrt{\frac{(\beta\gamma)_p}{(\beta\gamma)_{\text{ion}}}}\ \sigma_\theta(p)$$

which is **1** wherever the ion is γ-matched — the two lower configurations — and **√2** where it is rigidity-capped with A/Z = 2, i.e. ^{6}Li at the top. So a ^{6}Li carries the proton's divergence at 41 and 100 GeV and pays the √2 only at 18 × 275 — *not* a blanket factor, which is what an earlier draft of this programme's correction assumed.

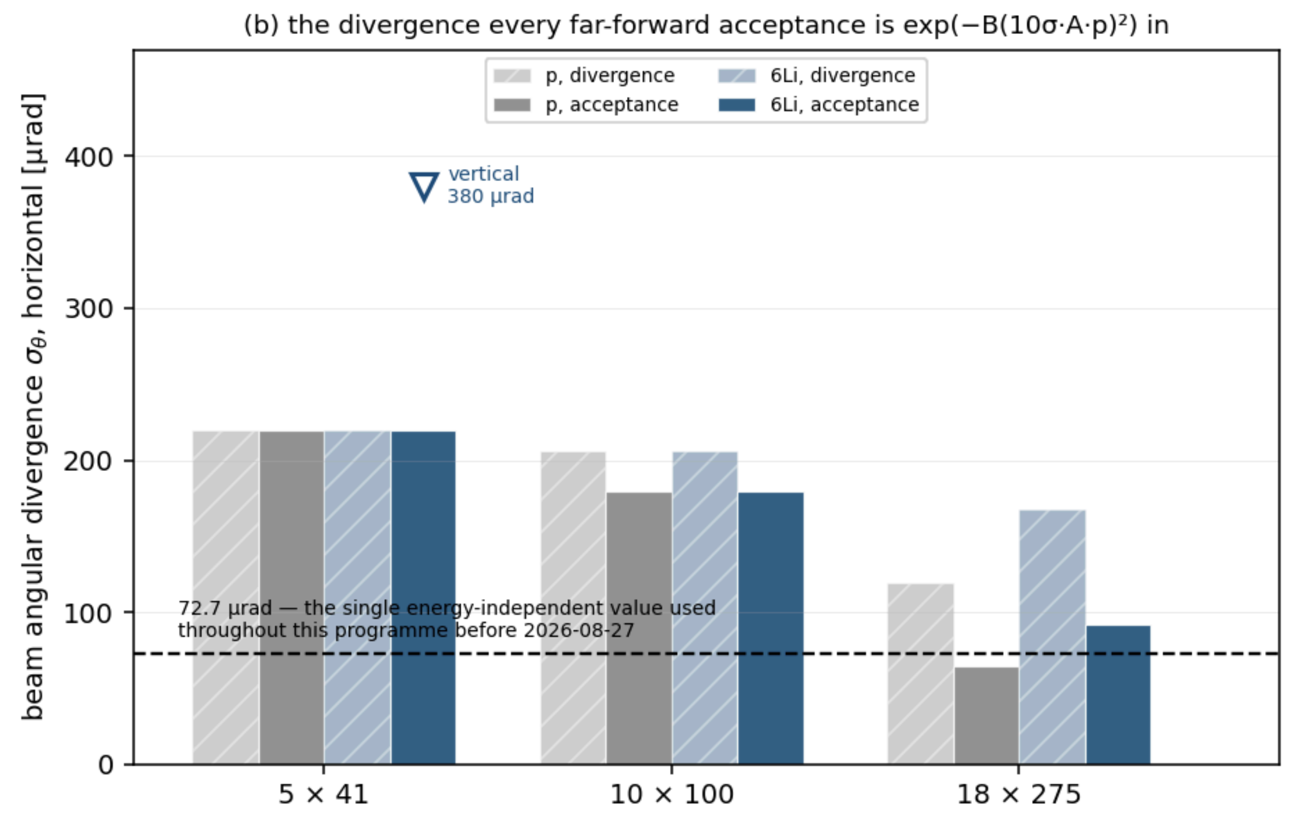

Figure 2 · The divergence. Protons and ^{6}Li per configuration and optics. The species √2 appears only at 18 × 275, where ^{6}Li is rigidity-capped; at the two lower configurations it is γ-matched and identical. The dashed line is the single energy-independent 72.7 μrad this programme used until 2026-08-27 — below almost everything. `eic_beam_figures.py`, `reco.sigma_theta_for`.

Is equal ε_N defensible for lithium? Gold is the published test. Scaling the 275 GeV proton to Au at 110 GeV/u by $1/\sqrt{(\beta\gamma)}$ alone predicts 236 μrad against an observed 218 (h) and 379 (v) — so $\varepsilon_N(\text{Au})/\varepsilon_N(p)$ = 0.85 horizontally and 2.6 vertically. Intrabeam scattering at fixed beam current goes as Z^3/A^2: **0.75 for ^{6}Li**, 1 for a proton, ≈17× that for gold. RHIC deuterons showed no measurable IBS growth while gold grew 20–45%/h. Lithium is a proton-class ion, so equal ε_N is the right starting assumption — and it is an assumption, not a specification.

5 · The ePIC detector

5.1 · Far-forward — where this programme lives

SUBSYSTEM	POSITION	ACCEPTANCE	TECHNOLOGY	STATUS
B0	`z ≈ 5.4–6.4 m`	`5.5 < θ < 20 mrad`	AC-LGAD + EMCal	design value

Off-Momentum Detectors	`z ≈ 22.5, 24.5 m`	`θ < 5 mrad, x_L 0.45–0.65`	AC-LGAD	design value
Roman Pots	`z = 32.55, 34.25 m`	`θ < 5 mrad, x_L 0.6–0.95,` **`inner edge optics-dependent`**	AC-LGAD, in-vacuum	geometry file (current `main`)
ZDC	`z ≈ 37.5 m`	`θ < 4–5.5 mrad, neutrals`	W/Si + crystal	design value

Table 4 — the Roman-Pot z positions are read from `compact/far_forward/roman_pots_eRD24_design.xml` on the current `main` branch and are **not** the 26 / 28 m the Yellow Report, the ePIC wiki and several papers still quote. The geometry moved; the documents did not.

The Roman-Pot sensor and its readout. AC-LGAD at **500 × 500 μm pitch** with a 100 × 100 μm metal electrode over a **30 μm active thickness**, bump-bonded to EICROC. Timing ≈35 ps per plane is a *requirement*; HPK AC-LGADs measure 20–35 ps in test beam. Spatial resolution ≈140 μm from the pitch alone; ≈20 μm has been shown with charge sharing, but ePIC's stated pot baseline is **binary 500 μm without charge sharing**. EICROC provides a 10-bit 25 ps time-of-arrival TDC and an **8-bit 40 MHz SAR ADC** for charge.

The pot insertion is quantised. Modules are 16 × 16 mm and the insertion is 2.4 cm + a per-energy offset (0.27 / 0.71 / 2.96 cm at 18 × 275 / 10 × 100 / 5 × 41), with the geometry file's own comment: *"These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."* The 1 mm aluminium RF shields are currently **commented out** — *"we don't know if we will even need it — it depends on the beam impedence + RF heating, etc. Oct. 2025"*.

Rates and environment. The ePIC Preliminary Design Report gives a few Hz per channel in normal running and *"30–50 kHz at ~10σ"* from beam halo, on STAR experience. In-vacuum cooling is ≈100 W per sensor layer. No published radiation fluence or dose for the pot station was found.

5.2 · Central detector

QUANTITY	VALUE	SOURCE	STATUS
EMCal σ_E/E, backward (η < −2)	`2%/√E ⊕ 1%`	YR requirement table	**requirement**
… backward transition (−2 < η < −1)	`7%/√E ⊕ 1%`	YR	requirement
… barrel (|η| < 1)	`10%/√E ⊕ 1%`	YR	requirement
… forward (η > 1)	`7%/√E ⊕ 1%`	YR	requirement
tracking σ_p/p	`√((a p)² + b²), a, b η-piecewise`	—	**PLACEHOLDER — no published source**
tracking σ_θ	`5/3/2/1/2/3/5 mrad by η`	—	**PLACEHOLDER**
electron ID efficiency	`0.70–0.95 by η`	ATHENA JINST 17 P10019 Table 5; ECCE NIM A 1055 168464	**proposal-era, not ePIC**

Table 5 — the calorimeter rows are Yellow Report *requirements*, which is what this programme uses and what it should say it uses. The tracking rows are stand-ins with no published source, flagged as such in `polligen/reco.py`'s own docstrings, and the electron-ID curve is interpolated between ATHENA and ECCE anchors because no official ePIC curve exists.

6 · What this programme assumes that the specification does not give it

Ranked by how much a wrong answer would move a published number.

#	WHAT IS MISSING	WHAT WE DO INSTEAD	OWNER
1	**Lithium beam divergence and Δp/p**, per configuration and cooling scenario	derive from the proton tables with the γ-matched species step (§4.1)	C-AD / ePIC FF WG
2	**Is there a high-acceptance optics at 41 GeV?** The YR's two rows are identical there	assume not; report 4 §7 shows the coherent channel depends on it	C-AD
3	**Lithium emittance, intensity, β* and luminosity** — nothing published	assume proton-class ε_N, defended by the gold calibration and the Z³/A² IBS scaling	C-AD / ANL source group

4	**EICROC input charge dynamic range in fC**, and LGAD gain suppression at ~9 MIP	assume the charge is usable; report 4 §4.1 shows the far-forward Z-ID depends on it entirely	OMEGA-IJCLab / BNL AC-LGAD group
5	**Per-optics transfer matrix at the pot planes** — R_{11}, R_{12}, R_{21}, R_{22}, D, D′	use R_{12} = 30.6 m, measured at 18 × 275 only, and work in angle elsewhere	ePIC FF WG
6	**ePIC tracking resolution and electron-ID** — no published curves	η-piecewise stand-ins, flagged in the code and in Table 5	ePIC
7	**Radiation environment at the Roman Pots** — no published fluence or dose	not modelled	ePIC FF WG
8	**Polarization-preservation scheme for ^{6}Li**, which cannot use Siberian snakes	assume it works at the EPIOS source targets	C-AD / EPIOS

The honest summary. The electron and proton beams are specified in detail and this programme uses them as such. The *ion* beam is specified for gold and, thinly, for ^{3}He; for lithium essentially nothing exists beyond a top energy and a G factor, and §4 is a derivation, not a citation. The detector is specified where it matters most for us — the far-forward geometry is in a file we can read — and least where the central reconstruction lives, which is why Table 5 carries two rows marked PLACEHOLDER rather than a number that would look like a specification.

References

[1] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (split across `refs/2103.05419_part1.pdf` … `part4.pdf`). Tables 10.1 and 10.2 (beam parameters per configuration and optics), Table 11.48 (Roman-Pot p_T smearing), §11.6 (far-forward), §14.5.3 (superconducting nanowires), and the detector requirement tables.

[2] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794 (`refs/2510.10794.pdf`). Table I (HERA vs EIC parameters); the species table with G and mc^2/G; the polarized-lithium case.

[3] E. Hamwi, G. H. Hoffstaetter, *Polarization Transmission in the Electron-Ion Collider's Hadron Storage Ring*, Phys. Rev. Accel. Beams 29 (2026) 073501, arXiv:2509.18558 (`refs/2509.18558.pdf`). Table I, the species-dependent resonance spectrum; the statement that small-G ions cannot use Siberian snakes.

[4] `eic/epic`, branch `main`: `compact/far_forward/roman_pots_eRD24_design.xml` and `compact/fields/beamline_*.xml`, read directly 2026-08-26. Station positions, module size, per-energy insertions, the RF-shield comment.

[5] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (`refs/ePIC_far_forward_talk_DIS_2023_v2.pdf`), and the ePIC Collaboration Meeting, JLab, July 2025. Acceptances, the AC-LGAD specification, the in-vacuum cooling load, the x-moving layer.

[6] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (`refs/2108.08314.pdf`). Table I, the far-forward acceptance windows this programme's router uses.

[7] ePIC Collaboration, *Preliminary Design Report*, September 2024. Roman-Pot rates: a few Hz per channel in normal running, 30–50 kHz at ~10σ from beam halo.

Appendix A · Revision history

DATE	REVISION
2026-08-27	First version. Written after the γ-matching correction (plans/10) made it clear that the programme had been carrying an accelerator assumption — that ion energies scale with rigidity — that no document supports and that the published gold and ^{3}He menus refute. The beam sections are the corrected picture; the detector sections separate requirement from placeholder; §6 is the list of what we are assuming on the machine's and the detector's behalf.

Status vocabulary: **design value** — appears in a design document as a parameter; **requirement** — a specification the subsystem must meet, not a demonstrated performance; **measured** — test beam or operations; **derived** — computed here from published inputs, with the derivation shown; **unknown** — not published anywhere we could reach. Beam divergences and Δp/p are Yellow Report Tables 10.1 and 10.2 unless stated. Ion energies are computed by `fastsim/polli_fastsim/beams.py` and pinned by `fastsim/tests/test_beams.py`. Built by `reports/build_report.py`; figures from `evgen/scripts/eic_beam_figures.py`.